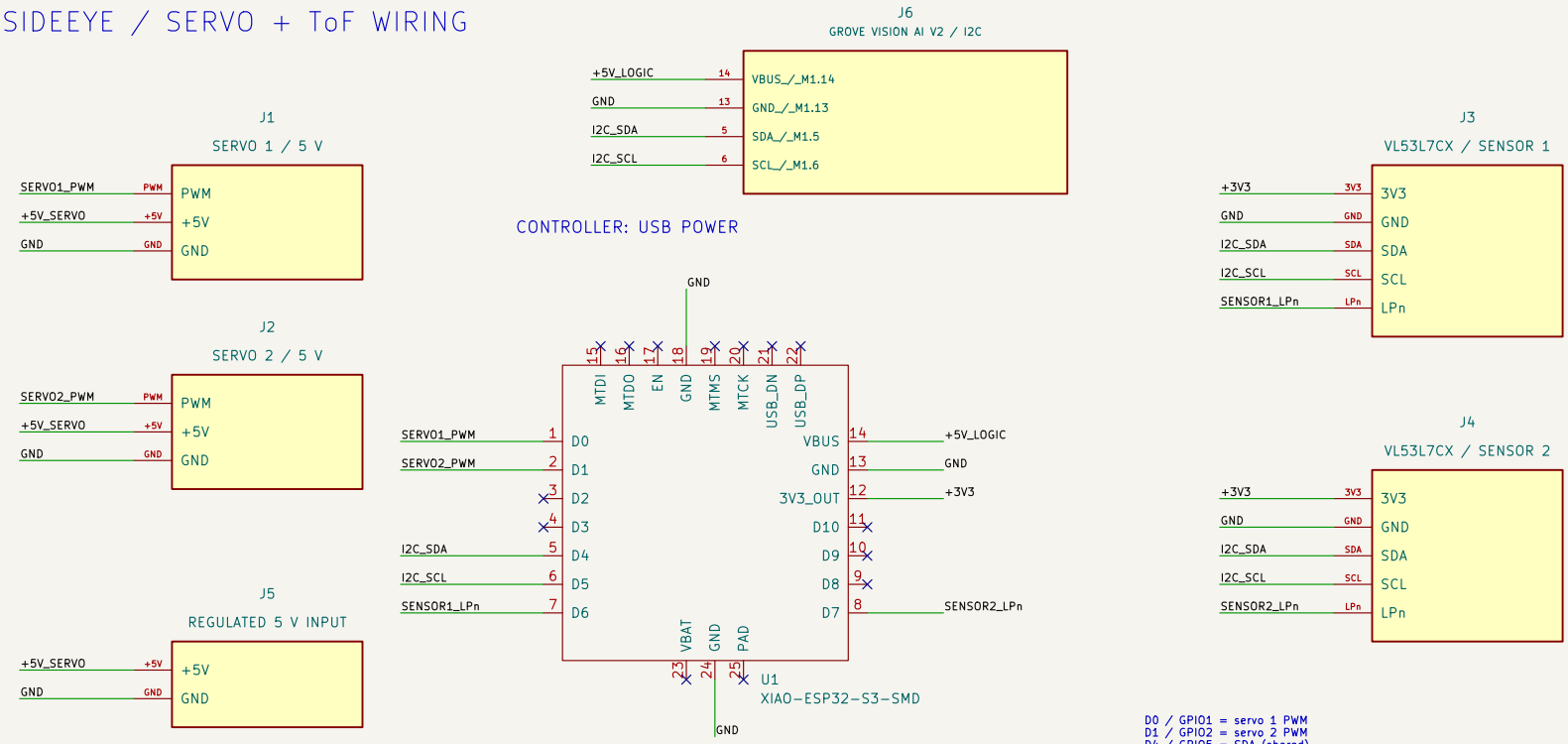
SIDEEYE / SERVO + ToF WIRING



D0 / GPIO1 = servo 1 PWM
D1 / GPIO2 = servo 2 PWM
D4 / GPIO5 = SDA (shared)
D5 / GPIO6 = SCL (shared)
D6 / GPIO43 = sensor 1 LPn
D7 / GPIO44 = sensor 2 LPn

WIRING NOTES

1. U1 is a XIAO ESP32-S3 host; Grove Vision AI V2 uses a separate Himax HX6538.
2. REQUIRED: selective adapter / separate wiring. Connect only M1 pins 5, 6, 13, 14 to XIAO.
   Do not fully stack the 14-pin socket: isolate all other contacts, including M1 pin 12 (3V3).
   D0/D1/D6/D7 go ONLY to servos / LPn; Grove D3 is RESET, D8 is SD/SPI clock.
3. Power XIAO by USB; its VBUS feeds Grove M1 pin 14. J5 powers servos; allow for stall current.
   All GNDs common. Servo +5V stays separate from logic power. Check USB supply capacity.
4. D4/D5 share I2C between Grove and both ToF boards; use 3.3 V pullups, account for all parallel pullups.
   Use the Grove XIAO socket SDA/SCL (3.3 V), not its separate Grove 4-pin port. Bus: 400 kHz maximum.
5. D6/D7 control sensor LPn separately. Enable one sensor at a time and assign unique addresses.
   Example 7-bit addresses: ToF 0x29 / 0x2A, Grove 0x62. ST driver may use shifted 8-bit addresses.
6. J1-J4 are functional interfaces: match labels to your breakout. Use 3.3 V-compatible ToF boards.
   I2C_RST low; INT unused (polling). Check XIAO 3V3 budget for both boards; PWM is 3.3 V logic.

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